

Zhiqiang Xu

Software Architect

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Work as a hacker. Hack as an artist.

SUMMARY

Bachelor's degree in Traffic Engineering with 19 years of experience in embedded software design and development across automotive, robotics, and mobile devices. Experienced in the full lifecycle of embedded system software development, from requirement analysis and system design to software implementation, testing, verification, and integration. Skilled in building workflows from scratch, simplifying processes, creating modular and scalable software architectures, and delivering high-quality solutions.

CORE SKILLS

- Embedded system software architecture design, development, debugging, and integration
- Expertise in Linux, C, Bash, Git, and Docker
- Efficient workflow improvement to enhance overall productivity
- Strong communication skills

WORK EXPERIENCE

- 2022.04 - Present **Senior Software Engineer, Zeekr Automotive, Shanghai**
- Led the design and development of cockpit SOA communication middleware.
 - Responsible for developing a tool for SOA communication matrix code framework conversion and supporting multi-platform SDK compilation and release.
 - Led the development of a Wireshark-based plugin for SOMEIP Ethernet message parsing, enhancing debugging efficiency.
 - Led the use of Docker to build a unified development environment, ensuring consistency between development and production environments.
 - Led the design and development of cockpit diagnostic middleware and services.
 - Responsible for developing the cockpit controller diagnostic upgrade module.
 - Led the design and implementation of a peripheral upgrade solution for the cockpit.
- 2019.10 - 2022.03 **Software Architect, SAIC Commercial Vehicle Technology Center, Shanghai**
- Responsible for revising and maintaining OTA protocols for the commercial vehicle IoT system.
 - Led the establishment of continuous integration services for the smart connected vehicle department.
 - Led PoC verification and bench setup for MTK cockpit solutions.
 - Responsible for the design and review of the serviceization of domain controller function signals via SOMEIP.
 - Led the design of the navigation synchronization solution for front and rear seats.
 - Led the design of a lightweight single-core multi-system solution based on containers and LXC.
- 2018.06 - 2019.09 **Senior Software Engineer, Shanghai Yunshen technology Co., Ltd. , Shanghai**
- Led the overall system architecture design and development of a service robot.
 - Responsible for developing a general-purpose real-time system based on CAN bus and RTThread.
 - Led the development of Docker-based ROS development, compilation, and runtime environments.
 - Led the completion of OTA functionality for the robot system.
 - Responsible for recruitment and technical business coordination.

- 2016.07 – 2018.05 **Software Solution Expert, Shanghai clever mrobot technologies Co., Ltd.**, Shanghai
- Led the development and architecture design of robot software systems.
 - Led BSP development for Android tablets based on RK3288 platform.
 - Led the optimization and design of flexible auto-upgrade architecture for ROS programs.
 - Led the unification of the development and runtime environment for ROS using Docker technology.
- 2010.06 – 2016.06 **Embedded Software Engineer, Shanghai Phicomm. Inc.**, Shanghai
- Responsible for the development of Android smart devices based on Marvell and Qualcomm chips.
 - Led fast installation and deployment configuration of Qualcomm compilation environments.
 - Led Dockerization of Android compilation environments to ensure consistency.
 - Responsible for developing sensor drivers and battery management systems.
- 2009.10 – 2010.06 **Embedded Software Engineer, Shanghai iBingo network technology Co., Ltd.**, Shanghai
- Responsible for developing dynamic effect themes for feature phones based on the MTK platform.
 - Developed multiple dynamic themes and screensavers.
 - Led the development of breakpoint-resumed download functionality for mobile phones.
- 2008.11 – 2009.10 **Embedded Software Engineer, Shanghai Dragontec Group**, Shanghai
- Responsible for developing iPod interface and applications for Cadillac's IVI systems.
 - Assisted in the porting of Hitachi's digital TV software from VxWorks to Linux.
 - Participated in the OpenVG porting project, responsible for unit testing and regression testing.

PROJECT EXPERIENCE

- 2023.09 – Present **EX1E/EX1H/DX1H, Zeekr Automotive**
- Project Background:** Zeekr 3.0 platform cockpit controller UXC (Qualcomm 8295)
- Responsibilities:** Led the design and development of diagnostic services and diagnostic upgrades for the 3.0 platform cockpit controller.
- Technical Contributions:**
- Developed the diagnostic middleware for the Zeekr 3.0 platform cockpit controller based on Hengrun UDS diagnostic protocol stack.
 - Used UML to design system architecture, component diagrams, and sequence diagrams for the middleware and completed the overall design.
 - Designed and developed a replacement tool for the original protocol stack vendor's diagnostic configuration system.
 - Responsible for the development and validation of various Basetech projects.
 - Developed diagnostic services for various vehicle models based on the self-developed diagnostic middleware platform SDK.
 - Designed and implemented the platformized upgrade solution for cockpit domain private nodes.
 - Led the complete optimization and reconstruction of the diagnostic upgrade module, adding parallel flashing capability, reducing the full upgrade time from 40 minutes to 15 minutes.
- Achievements:** Successfully implemented a platformized cockpit diagnostic development framework. By encapsulating third-party diagnostic protocol stacks, it provides protocol-stack-independent diagnostic development interfaces for functional developers, significantly reducing the labor required for diagnostic service development. The introduction of the replacement tool further decreased the dependency on vendor configuration tool licensing. The optimized diagnostic upgrade module embodied a high cohesion, low coupling design philosophy, reducing failure rates and improving upgrade efficiency.

2022.04 – 2023.10 **CS1E/CM2E/DC1E, Zeekr Automotive**

Project Background: Zeekr 2.0 platform cockpit controller (Qualcomm 8295)

Responsibilities: Led the system and software architecture design for the SOA communication platform.

Technical Contributions:

- Developed the SOA communication middleware that supports multiple operating systems (Android, QNX, Linux) based on open-source vsomeip and commonapi.
- Used UML to design system architecture diagrams, component diagrams, use case diagrams, and sequence diagrams for the middleware and completed the overall design.
- Built a unified development environment using Docker to ensure a consistent development experience.
- Implemented a simulation of service cross-domain communication on a single machine using Docker.
- Led the design and development of a Wireshark-based SOMEIP data packet parsing plugin, significantly improving development and testing efficiency.

Achievements: The self-developed SOA communication platform was successfully applied to Zeekr 2.x platform vehicle models. Its solid design, combined with supporting self-developed tools, enabled the efficient development and integration of smart driving-related services. In subsequent maintenance, the system proved to have excellent modularity and high maintainability.

2019.10 – 2022.03 **MIFA, SAIC Commercial Vehicle Technology Center**

Project Background: Qualcomm 8155 intelligent cockpit project, based on dual operating systems QNX and Android, and Linux single operating system, for the design and development of CAN signals and SOA services.

Responsibilities: Responsible for CAN signal interface design, SOA services matrix definition, and related feature implementation.

Technical Contributions:

- Revised and maintained the OTA air protocol.
- Managed code with LDAP-based access control and built the CI/CD system.
- Designed a container-based single-core multi-system solution.
- Led the architecture design for the implementation of front and rear seat navigation synchronization.
- Led technical discussions and drove the implementation of high-precision map functionalities.
- Responsible for the design and review of inter-domain Ethernet SOA services.

Achievements: Successfully delivered the designed functionality on schedule, with the project being completed on time.

2018.06 – 2019.09 **Astrid Robot, Shanghai Yunshen Technology Co., Ltd.**

Project Background: Development of service robots for the furniture retail and supermarket scenarios.

Responsibilities: Led the system architecture design and implementation of the robot's overall control solution.

Technical Contributions:

- Completed the platformized development framework for the general-purpose real-time system based on RT-Thread.
- Developed CAN driver support for STM32 in RT-Thread.
- Used UML to design the overall control solution for the robot, including system components, use cases, component activities, and sequence diagrams.
- Designed and implemented a sensor node system based on CAN bus, enabling real-time data transmission and processing from multiple sensors.
- Deployed the CI/CD system using Jenkins and Gerrit.
- Built a unified development and testing environment using Docker.
- Implemented remote firmware upgrade capabilities for sensors, reducing hardware maintenance costs.
- Built Docker runtime images for ROS, improving debugging and feature deployment efficiency.
- Used a virtual machine solution to configure and create images for the robot's central control computer operating system, improving consistency and reducing operational risks.
- Integrated Intel RealSense drivers into the Docker image, resolving hot-plug issues with cameras, and fully implemented Docker image-based operation for ROS.
- Ported a facial recognition algorithm and implemented a ROS node.

Achievements: Played a key role in the overall development of the Da Tian Shi robot, overcoming various technical challenges, and ultimately enabling OTA upgrades for the robot system. This provided the robot with the ability for online updates and maintenance, improving post-launch support.

- 2016.07 – 2018.05 **Noah Robot, Shanghai Mrobot Technologies Co., Ltd.**
Project Background: Development of logistics robots for hospital scenarios.
Responsibilities: Responsible for integrating the latest algorithms developed by each function team, preparing POC demonstration prototypes, investigating and resolving on-site technical issues, and designing the next-generation control system.
Technical Contributions:
 - Optimized the startup time and power management for Android tablets based on the RK3288 platform.
 - Designed and implemented the robot control system architecture based on ROS.
 - Used Docker images to encapsulate the system environment for consistency and repeatability, achieving success.
 - Standardized the compilation and packaging process for ROS programs and implemented USB-based auto-upgrade functionality.**Achievements:** Successfully delivered the project, greatly improving the stability and performance of the robot control system.
- 2010.06 – 2016.06 **C630/C832/F320/P680, Shanghai Phicomm Technology Co., Ltd.**
Project Background: Phicomm's proprietary brand of smartphone development.
Responsibilities: Responsible for device driver porting and tuning, and the development of the mobile device power management system.
Technical Contributions:
 - Developed automation tools to achieve fast installation and configuration of Qualcomm compilation environments, eliminating the need for complex Qualcomm development environment setup.
 - Dockerized the compilation environment to ensure consistency and reduce risks.
 - Introduced distcc and ccache to enable distributed and cached Android compilation, greatly enhancing Android release efficiency.
 - Designed and implemented a data modeling tool for SOC battery drivers, and significantly improved the user experience by optimizing the system's charge and discharge strategy.
 - Implemented multi-channel charging functionality based on SMB358.
 - Designed and implemented an efficient sensor driver development scheme, improving driver porting and tuning efficiency by over 6 times.**Achievements:** Through the accumulation and summary of development methods and practices, improved driver development and tuning efficiency, while also significantly enhancing the efficiency of version regression verification for compiled builds.
- 2009.10 – 2010.06 **iBingo iShow Theme System, Shanghai iBingo Network Technology Co., Ltd.**
Project Background: Developed dynamic effect systems for feature phones based on MTK chipsets.
Responsibilities: Responsible for developing dynamic themes and screensavers for feature phones based on a graphic development library.
Technical Contributions:
 - Ported the Allegro graphics library and its graphical algorithms.
 - Developed multiple dynamic theme desktops.
 - Developed a dynamic screensaver with an infinitely changing color ribbon effect.
 - Developed a dynamic landscape screensaver that switches between day and night.
 - Designed and implemented a templated system, enabling the automatic generation of themes by simply swapping graphic assets for the same theme type.
 - Responsible for the development of the download program's breakpoint-resumed download feature for mobile phones and created an automated publishing tool.
 - Developed a company promotional CD using Actionscript.**Achievements:** In under a year, from the company's inception, successfully developed multiple theme desktops and screensaver systems, along with production and publishing tools.
- 2008.11 – 2009.10 **'08, '09 Cadillac CTS HU Development, Shanghai Dragontec Group**
Project Background: Outsourced to Hitachi for collaboration on automotive development.
Responsibilities: Responsible for the development of the iPod interface and applications for Cadillac's IVI systems.
Technical Contributions:
 - Improved the algorithm for reading iPod playlists, reducing the time from 20 seconds to 0.5 seconds, significantly enhancing the user experience.
 - Developed an audio file header information parsing library, enabling the extraction of decoding parameters from audio files of different formats and providing them to the decoder, ensuring correct playback of various audio formats.**Achievements:** Delivered the project on time and gained extensive experience from the development of the iPod functionality and audio file header parsing library.

HOBBIES

Sports F1, Running, Skateboarding

DIY Mechanical Keyboards, Electronic
Gadgets

Movies Documentaries, Sci-Fi Movies

Reading Technical Books, Novels

GRANTED PATENTS

201510477097.X Login authentication method and system
201510612722.7 System for expanding operational capability of mobile terminal
201410613796.8 Mobile terminal and achievement method of virtual drive of same
201210368281.7 Headphone interface device and control method based on the headphone interface device
201510745100.1 A kind of key encryption method and system, electronic equipment
201210585845.2 Hardware and firmware independently updating system and method thereof
201420615063.3 A kind of mobile phone water enter preventing device
CN201811304800.7A Generation method of thermodynamic diagram of active area and server

MISCELLANEOUS

Contributed 3rd party projects **github**

- twip: Added proxy support.
- ChinaDNS-C: Fixed bugs and ported to tomato.
- Koreader: Fixed bugs and optimized compilation speed.
- tmk_keyboard: Added bluetooth support for usb2usb.
- RT Thread: Fixed STM32F1 bsp bugs. Wrote CAN driver for Stm32f4 using hal library.
- ChameleonMini-rebooted: Added NTAG-213, 215, 216 support.
- lsp-sonarlint: An Emacs plugin from using sonarlint as a lsp backend to help checking codes rules. Added c/c++ checking support.

My projects **github**

- vim_configs: Vim configure used for co-workers.
- battery_analyzer: A tool that auto sampling and use spline method to generate SOC table for battery driver
- csr_tool: Script which can dump firmware from CSR chips.
- Ultramanmedal: A lua script which can generate tags for zeta ultraman.
- bin2elf: Add elf header for bin file.
- wintoolset: An environment for running script tools under windows.
- trc2asc: Convert trc file which is saved by peak apps to asc file.
- asc2blf: Convert ASC to blf or blf to asc.
- exportdrawio: Export png svg pdf files from a drawio file.
- crackmfkey: Auto download sniff data from chameleon and calculate keys out.